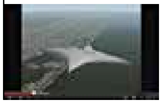

The Flying Shuriken

Watch this video to know exactly how the shuriken shaped aircraft (SBiDir-FW) flies: **http://dgit.in/RdeyJc**

bility issues pretty well (though it does not add much in the name of futuristic looks, and loses out on drool points).

Our next design - the Double Bubble D3 has taken a different direction, in terms of employing the latest in noise reduction strategies. An aircraft's noise might have two sources - the aerodynamic fluctuations by the flight, or the mechanical noise from the engines itself. What this design hopes to achieve is that the craft has the ability to stabilise itself by design, and not by power, which results in lesser air resistance and therefore, reduces on the aerodynamic fluctuations. It also manages to further channelise sound by placing a spherical plate to guard the underside of the engine, which reflects the mechanical noise straight back up to the heavens, because hey, the big guy upstairs doesn't care.

The boxed winged concept

Now, for some fantasy talks. Apart from simply making aircraft look suitably futuristic, plans are in motion to give them some mind blowing capabilities too. How about a plane that can mutate its shape during flight? Fans of anime will instantly recognize the image to closely resemble the shuriken. A team from the University of Miami have created this idea to bridge the compromise that is flying supersonic. The key here is the wingspan. Subsonic flight requires better wings to ensure efficient output from the power being pumped from the engines for the push. But, things change radically when something crosses Mach-1. Aerodynamics at it again, now the wings need to be as subdued as possible (along with everything else), to ensure stability during the flight. Up until now, designs were hard-built as a mixture of them both. But this new design, which has been given a nod (and some $100,000 in grant) by NASA as part of it's Innovative Advanced Concepts (NIAC) design

MIT's Double Bubble D8 design

competition, is a completely new take on the way in which the airborne vehicle should transform to adapt to the change in supersonic to subsonic speeds, and vice-versa. The radical part of the idea is that the aircraft will take off and land in one direction, while its flight will logically be at in a direction perpendicular to that. The supersonic bi-directional flying wing (SBiDir-FW) aircraft would rotate into supersonic configuration by folding winglets attached to the end of the wings in subsonic configuration. Folding them up again would see the aircraft rotate back again to subsonic orientation once again. The engine pod on the midsection would also be rotated in the opposite direction when switching modes, so that the orientation of the engine remains the same during the transition, in order to conserve the kinetic energy of the plane.

Tying up loose ends, we'd like to touch upon something that is probably at the undertone of it all. There's a strange sense of homecoming in all of these designs. When you take a close, analytical look at the designs they seem, in some way or the other, to be trying to model the natural

You'll be riding Ninja stars soon

practitioners of flight more and more closely. Birds, bugs, whatever (and we sometimes imagine dinosaur in there too), but finally the world agrees that the best way to do it is the way that it has been done for thousands of years. What should have been obvious is now in the fore. There's a strange sense of homecoming to the future, when you think about it.